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# Measuring fuel connectivity in coastal mallee shrublands using terrestrial mobile LiDAR

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## Abstract

**Background** Fuel metric derived from mobile laser scanners (MLS) have been used in forested areas to quantify fuel connectivity (or ladder fuels) and have potential to be applied to low woodlands and shrublands. Connectivity of fuel is important in shrublands as it relates to go/no-go and onset of crown fire thresholds. Coastal mallee is an exemplar of a fire-prone sclerophyllous shrubland found in Mediterranean climates and is important to characterize for potential thresholds to ignition, fire spread, and crown fire. Existing wildfire models developed for similar fuels rely on canopy height and cover; however, previous research suggests additional MLS-derived vertical and horizontal connectivity metrics may supplement and improve understanding of fire behavior in these fire-prone shrublands.

**Results** Point clouds were created from MLS surveys of 25 coastal mallee plots in South Australia. We compared strata-derived metrics, horizontal fuel metrics, and vertical fuel metrics by examining the correlation between metrics and with simpler canopy height measurements. We compared the mean and variance between 10 × 10 m subplots and the whole plot for each metric using Welch's *t*-test and Fligner–Killeen test, respectively. We also test each metric's ability to discriminate between age-rainfall classes and geographic regions using Kruskal–Wallis rank sum test. The height of the first and second strata together with the maximum vertical strata gap were the most useful strata-derived fuel metrics. They were not correlated with other simpler height metrics nor with each other and were significantly different between age-rainfall classes and geographic regions. Overstory cover is the preferred horizontal fuel metric over total vegetation cover above 0.5 m as the median overstory cover values were significantly different between geographic regions and had a wider range of values. Canopy height, height skewness, gap fraction, and vertical complexity index performed the best for low correlation with other metrics, insensitivity to plot size and ability to discriminate between age-rainfall classes and geographic regions.

**Conclusions** MLS-derived fuel metrics provide a robust method of measuring horizontal and vertical fuel connectivity in coastal mallee shrublands. These metrics can be used in studies of fire behavior in Mediterranean climate shrublands to improve the understanding of ladder fuels and fuel structure in fire behavior of these fire-prone ecosystems. These metrics improve on current fuel assessment methods in objectivity, repeatability, and sampling time.


**Keywords** LiDAR, Mobile laser scanner, Mallee, Shrubland, Fuel metrics

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## Resumen

**Antecedentes** Las métricas derivadas del uso de escáner laser móviles (MLS) han sido usadas en áreas forestadas para cuantificar la conectividad de combustibles (o escaleras de combustibles) y tienen el potencial para ser aplicadas en matorrales o bosquecillos bajos. La conectividad de los combustibles es importante en matorrales, pues dado que relaciona el *va/ no va* (*go/no-go*) y determina el comienzo de los límites de los fuegos de copas. Los matorrales costeros de eucalyptus (*Mallee* de Australia) son un ejemplo de los matorrales esclerófilos propensos al fuego en ecosistemas mediterráneos y son importantes para caracterizar los límites potenciales para la ignición, la propagación del fuego y los fuegos de copas. Los modelos de fuegos existentes desarrollados para combustibles similares se basan en la altura del dosel y cobertura; sin embargo investigaciones previas sugieren que métricas adicionales derivadas de conectividades horizontales y verticales medidas con MLS pueden suplementar y mejorar el entendimiento del comportamiento del fuego en esos matorrales propensos al fuego.

**Resultados** Nubes de puntos fueron creadas mediante el uso de relevamientos con MLS en 25 parcelas de matorrales de Mallee en el sur de Australia. Comparamos métricas derivadas de estratos, métricas horizontales y verticales mediante el examen de correlaciones entre métricas y mediciones simples de altura del dosel. Comparamos la media y la varianza entre sub parcelas de 10 × 10 m y la parcela completa para cada métrica usando el test de Welch's t- y Fligner-Killeen, respectivamente. Probamos asimismo la habilidad de cada métrica para discriminar entre clases de edad y lluvias y regiones geográficas usando el test de suma de rangos de Kruskal–Wallis. La altura del primer y segundo estrato junto con el estrato vertical máximo de un abra del bosque fueron los estratos más útiles para las métricas derivadas. No estuvieron correlacionadas con otras métricas simples ni tampoco con las otras métricas y fueron significativamente diferentes entre clases de edad - lluvias y regiones geográficas. La cobertura del estrato superior fue la métrica preferida sobre la vegetación total por encima de los 0,5 m mientras que en los valores de mediana, la cobertura fue significativamente diferente entre regiones geográficas y tuvo valores mucho más amplios. La altura del dosel, la desviación de la altura, las fracciones de claros y el índice de complejidad vertical fueron los que obtuvieron las mejores performances para correlaciones bajas con otras métricas, insensibilidad al tamaño de parcela y la habilidad para discriminar entre las clases de edad - lluvias y regiones geográficas.

**Conclusiones** Las métricas derivadas de MLS proveen de un método robusto para medir la conectividad horizontal y vertical del combustible en matorrales costeros de Mallee. Estas métricas pueden ser usadas en estudios de comportamiento del fuego en matorrales de clima mediterráneo para mejorar el conocimiento de las escaleras de combustible y estructuras de combustibles en estudios de comportamiento del fuego en ecosistemas proclives al fuego. Esas métricas mejoran la determinación de métodos corrientes de determinación de combustibles en cuanto a objetividad, repetición y tiempos de muestreo.


## Background

Shrublands are an important physiognomic type covering about 13% of the Earth's land area (Peng et al. 2021), and wildfires in semi-arid shrublands are renowned for causing devastating impacts to humans and ecosystems. Shrublands in areas such as California (Keeley et al. 2009, 2004; Keeley and Syphard 2019; Weise et al. 2005), the European Mediterranean region (Sağlam et al. 2008; Papió and Trabaud 1990; Fernandes et al. 2000), Chile (Castillo S et al. 2020), South Africa (Van Wilgen et al. 1985), and Southern Australia (Filkov et al. 2020; Enright et al. 2012; Fontaine et al. 2012) are considered fire-prone due to the relatively short but continuous fuel structure and dry flammable fuels due to the semi-arid climate. Modeling wildfire spread in shrublands within semi-arid and Mediterranean climates is difficult due to the complex nature of the fuel bed structure typically found in these ecosystems (Anderson et al. 2015). While shrublands can exhibit extreme fire behavior, they are also known to be barriers to fire spread under moderate or prescribed burning conditions partly due to the complex arrangement of fuels, fuel moisture, and environmental thresholds (Anderson and Anderson 2010; Cruz et al. 2010, 2013; Stephens et al. 2008; Weise et al. 2005). To better understand fire behavior in shrublands, improved measurements of fuel complexity are required, as most existing shrubland fire models rely on simplistic estimates of fuel (Telfer et al. 2024).

A recent survey of researchers and fire practitioners found that measuring the horizontal and vertical connectivity of wildland fuels is key to improving understanding of fire spread in a shrubland vegetation type (Telfer et al. 2024). Ladder fuels are critical components in fire behavior; by bridging the vertical gap between surface litter and the canopy, ladder fuels can play a significant role in influencing flame height and the probability of crown fires. Measuring vertical connectivity and ladder fuels in shrublands using traditional in situ fuel assessment

methods has proven to be time-consuming and challenging. Methods such as point-intercept transects are time-consuming, as transporting equipment such as range poles and tape measures is cumbersome in the short dense shrubland vegetation. They are also prone to observer bias and measurement errors when readings are difficult to obtain at or above eye level. Alternatives to detailed in situ fuel methods are estimates using fuel hazard guides (Hines et al. 2010; Gould, McCaw, and Cheney 2011; Cheney et al. 2012). These guides were primarily designed for assessing fuel in forested areas for specific purposes, and while some guides have adapted this method to shrublands, they have not been validated against objective ground-truthed methods (DENR 2012).

Advances in 3D remote sensing technologies, such as Light Detection and Ranging (LiDAR) point clouds, have proven accurate (Sexton et al. 2009) and facilitated the development of new methodologies for classifying and measuring fuel (Holt, Marre, and Telfer 2021; Hillman et al. 2021a, 2021b, 2021c; Chen et al. 2016; Andersen et al. 2005). LiDAR-derived metrics are also objective and repeatable, addressing concerns raised about visual fuel estimate methods (Watson et al. 2012; Spits et al. 2017). LiDAR-derived metrics allow quantification of more complex structural characteristics of shrublands which are not easily measured with traditional field techniques (Kramer et al. 2014; Alonso-Rego et al. 2020).

LiDAR-derived metrics also address ambiguity around the definition of strata introduced by fuel hazard guides commonly used in Australia (Hines et al. 2010; DENR 2012; Cruz 2024). For example, metrics calculated for strata derived from the inflection point method (Wilkes et al. 2016) use a data-driven approach to quantifying fuel strata. This method does not presume an existing number or distribution of strata, removing observer bias and subjectivity from fuel metrics. Ocular methods currently employed using the South Australian Overall Fuel Hazard Guide are adapted from taller forest systems, providing only approximate guidance for "near-surface" and "elevated" fuel structures. Due to the dense, complex, and almost vertically continuous nature of coastal mallee fuel, it is difficult to visually distinguish fuel strata; thus, a data-driven approach is desirable to have an objective, repeatable method of assessing fuel structure. Destructive fuel assessment methods are laborious and time-consuming due to the dense nature of the vegetation.

In comparison with the South Australian Overall Fuel Hazard Guide, several limitations of LiDAR methods can be highlighted, particularly in estimating surface litter and bark fuels. Separating surface litter from the ground and near-surface layers in discrete-return LiDAR point cloud has been shown to be difficult (Wallace et al. 2022) and bark fuels have yet to be quantified using LiDAR methods (Hillman et al. 2021a, 2021b, 2021c; Chen et al. 2016). As with most remote sensing methods, occlusion in dense vegetation is a known issue for both airborne and terrestrial LiDAR (Wilkes et al. 2017; Campbell et al. 2018).

LiDAR-derived metrics are increasingly used to quantify various aspects of fuel—including canopy structure (Ross et al. 2022; Hillman et al. 2021a, 2021b, 2021c; Engelstad et al. 2019; González-Ferreiro et al. 2014; Erdody and Moskal 2010), fuel biomass (Wallace et al. 2017; Chen et al. 2017; Jakubowksi et al. 2013; Hopkinson et al. 2004), overall fuel distribution (Skowronski et al. 2016; Chen et al. 2016; Hermosilla et al. 2014), and structure (Olszewski and Bailey 2022; Viedma et al. 2024). High-density point clouds from MLS and terrestrial laser scanners (TLS) overcome some limitations of above-canopy airborne laser scanners (ALS) such as occlusion of subcanopy fuel layers (Ruhan et al. 2023; Tiede et al. 2024) and enable detailed structural measurements of surface and ladder fuels. MLS point clouds have been shown to be comparable to TLS (Levick et al. 2021; Spadavecchia et al. 2023; Zhong et al. 2017; Liang et al. 2014). Simultaneous Localization and Mapping (SLAM) based MLS also provides advantages over TLS, including reduced capture time (Donager et al. 2021; Holvoet et al. 2025), reduced processing time, and reduced occlusion by customizing capture paths (Levick et al. 2021; Liang et al. 2014; Tiede et al. 2024; Donager et al. 2021). These advantages allow larger plots to be captured at high resolution particularly as MLS technology improves (Scaioni et al. 2025). MLS shares occlusion limitations with ALS and TLS when vegetation is dense and a suitable path cannot be navigated, difficulty in distinguishing bark fuels and limitations in determining litter depth (Chen et al. 2016).

LiDAR point cloud metrics that have been used to measure fuel distribution and structure in tall forested areas (Olszewski and Bailey 2022; Ross et al. 2024; Viedma et al. 2024) are now being explored for application in shrublands. Shrublands have relatively short gaps between surface fuels and the canopy, requiring precise measurements to quantify connectivity (Catchpole 1987). The high density of MLS point cloud (e.g., more than 100 points/$m^3$) has potential to provide accurate assessment of ladder fuels in shrublands and can improve our understanding of fire spread, prescribed burning thresholds, and aid in risk management.

### Coastal mallee shrublands

Mallee shrublands are a fire-prone vegetation association characterized by an overstory of multi-stemmed eucalypts found in semi-arid and arid regions of Southern Australia. Classification into broad shrubland, woodland, or forest fuel types (Hollis et al. 2015) is difficult due to the

dense shrubby understory (suggesting a shrubland) and relatively short eucalypt-dominated overstory (2–6 m) (suggesting a woodland or forest), which may be discrete from the shrub understory or completely integrated into a single fuel layer depending on time since fire and biogeographical constraints (Hill 1989). Fire and land managers often will consider mallee ecosystems a separate fuel type with their own fire characteristics (Kenny et al. 2024; DENR 2012; Weir 2018). Wildfires in mallee vegetation are renowned for their intensity and rapid rate of spread under dry and windy conditions (Filkov et al. 2020; Cruz et al. 2010; McCaw 1997; Bradstock and Gill 1993).

The relatively short canopy height of mallee ecosystems contributes to abrupt increase in fire behavior (the so-called go/no-go threshold) with small changes in environmental conditions (Cruz et al. 2013). Under mild weather conditions, such as those desired for prescribed burning, mallee shrublands often fail to sustain active fire spread due to the discontinuous nature of the fuels (Telfer et al. 2024; Bradstock and Gill 1993). This apparent fire spread threshold poses challenges for fire managers attempting to mitigate the risk of large extensive wildfire through prescribed burning. "Unbounded prescribed burning" (Gill 2012) commonly used in mallee vegetation, while effective in reducing the spread of subsequent wildfires (Fernández-Guisuraga and Fernandes 2024), has an inherent level of risk (Sandell et al. 2006; Tolhurst 2007). This risk can be reduced through effective use of predictive fire behavior models and prescriptions.

In this research, we examine a specific mallee ecosystem known as coastal mallee. Coastal mallee is predominantly found within 30 km of coastal regions of Southern Australia (Berkinshaw 2010), with higher annual average rainfall (450–800 mm) when compared with semi-arid and arid mallee (200–450 mm) ecosystems of Southern Australia (Cruz et al. 2010; DoEE 2017; Bureau of Meteorology 2020). This rainfall isohyet distinction is arbitrary and there is likely overlap between coastal and semi-arid mallee in terms of rainfall, fuel structure, and fire behavior. However, when compared with semi-arid mallee, coastal mallee is considered more continuous fuel (Telfer et al. 2024) and does not have a specific empirical model for predicting fire spread thresholds.

The prescriptions used for burning in coastal mallee in South Australia rely on the Catchpole et al. (1998) model and estimates of "heathland height" (Marsden-Smedley 2011). The semi-arid mallee rate of spread model (Cruz et al. 2013) presents an alternative for predicting fire behavior in coastal mallee. However, the overstory cover and horizontal connectivity of fuel exceed the bounds of the input parameters and do not meet the assumption of a horizontally discontinuous fuel array. Land managers (including the authors' own experience with prescribed fire) have found these prescriptions and models to be unreliable in coastal mallee and instead use anecdotal evidence and previous experience to estimate the best conditions to conduct burns, in part due to the variability in fuel structure of coastal mallee (Aaron Macumber, South Australian National Parks and Wildlife Service, Port Lincoln, South Australia, personal communication).

In this study, we aim to explore which LiDAR-derived metrics are most important and explanatory for measuring fuel structure in coastal mallee shrublands of South Australia using very high-density MLS point clouds. Our analysis uses standard structural metrics (e.g., height-based metrics, structural diversity, cover) commonly used in ecology and forestry applications along with vertical gap metrics derived from automatic vegetation strata detection (e.g., gap between strata, depth of strata) at a whole-plot (0.36–1 ha) and 10×10 m subplot scale. By quantifying these complex fuel arrangements characteristic of these fire-prone ecosystems, we aim to narrow the list of potential LiDAR-derived metrics to determine which metrics explain the fuel structural diversity found within coastal mallee shrublands across a range of age classes and biogeographical regions. Being able to explicitly quantify and map fuel structure, including ladder fuels, will assist in understanding fire behavior in coastal mallee shrublands and potentially improve fire models for prescribed burning and wildfire predictions. Metrics derived for coastal mallee may also have application in other shrublands around the world, as they are representative of fuel complexity found in other semi-arid and Mediterranean shrublands.

## Methods and materials

### Description of study area and sites

Study sites were chosen from planned prescribed burns in coastal mallee on Kangaroo Island and Eyre Peninsula in South Australia (Fig. 1) to capture the diversity of fuel structure in coastal mallee. Fuel bed structure ranged from shorter height (1.5–6 m) in Port Lincoln National Park (Fig. 2a and b) through to taller (3–9 m) and denser canopy cover on Kangaroo Island (Fig. 2c and d). Sites on the Eyre Peninsula sites were located in Murrantata Conservation Park, Lincoln National Park, and surrounding contiguous coastal mallee. The Kangaroo Island sites were located on public land reserves on Kangaroo Island, with the most westerly site being in Seal Bay Conservation Park and the most easterly in Dudley Conservation Park. The dominant overstory consisted of mallee eucalypt species including *Eucalyptus diversifolia*, *E. remota*, *E. albopurpurea*, *E. conglobata*, *E. rugosa*, *E. phenax*, *E. angulosa*, and *E. oleosa* ranging in height from 2 to 10 m. The understory consists of more than 150 shrub species with heights

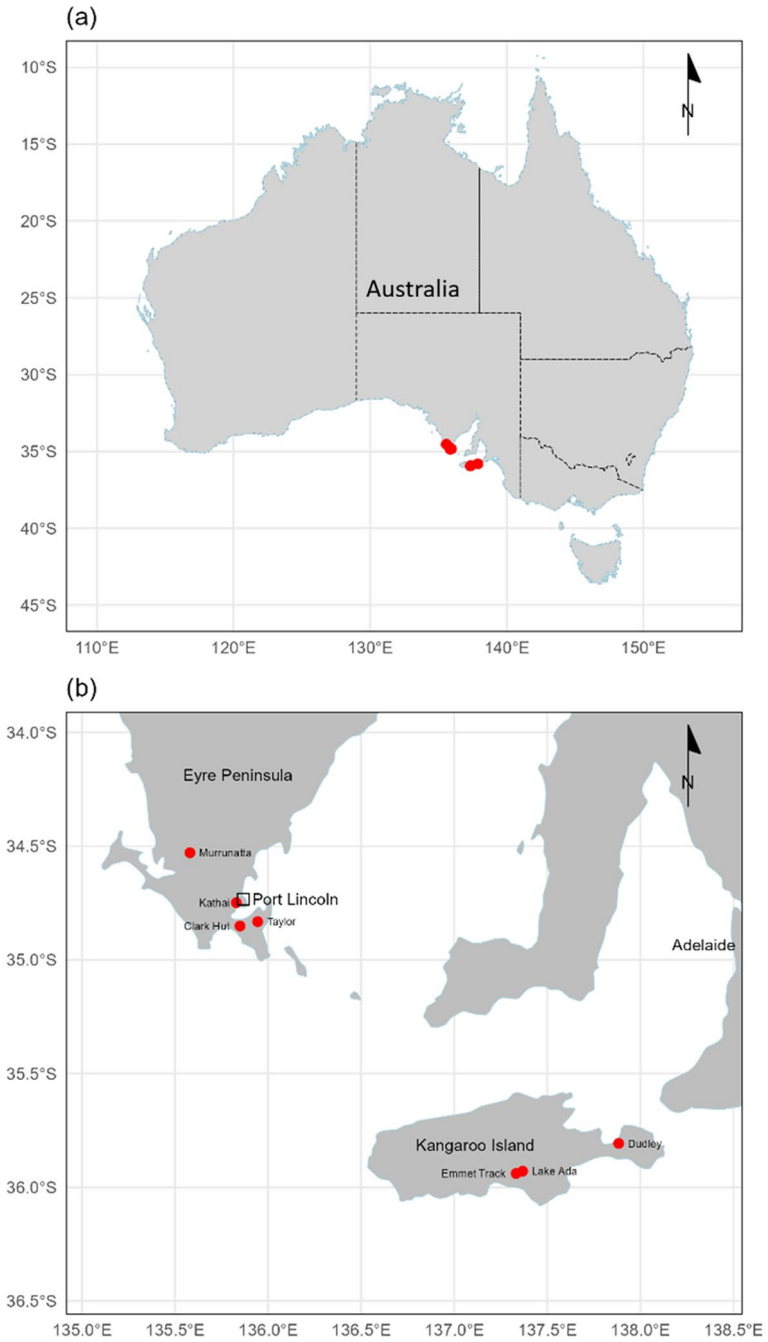



**Fig. 1** Study site location of coastal mallee in South Australia (**a**) and detailed location of sites on Eyre Peninsula and Kangaroo Island (**b**)

ranging from 0.2 to 3 m with dominant species including *Acacia* spp., *Hakea* spp., *Melaleuca* spp., *Banksia* spp., and *Xanthorrhoea* spp. (Berkinshaw 2010). Within the study sites, 25 plots were selected for fuel structure sampling. Sites had fuel age of 14 to 57 years since last fire. Average annual rainfall ranged from approximately 460 to 550 mm (Bureau of Meteorology 2020). Sites were grouped into 4 classes: fuel age less than 30 years and average annual rainfall (AAR) less than 500 mm, fuel age less than 30 years and AAR equal to 500 mm, fuel age less than 30 years and AAR greater than 500 mm, and fuel age greater than 30 years (AAR not considered due to low sample size).




**Fig. 2** Range of fuel height and density of coastal mallee sites where **a** is an aerial image (approx. 25% canopy cover, 90% total cover) and **b** is a terrestrial oblique image of coastal mallee vegetation (2–3 m tall) in Port Lincoln National Park. **c** is an aerial image (approx. 50% canopy cover and 95% total cover) and **d** is a terrestrial image of coastal mallee vegetation (7–9 m tall) on Kangaroo Island

### Data collection and preprocessing

Plots were sampled using a backpack-mounted Emesent Hovermap HVM-100 or ST-X (Emesent Pty Ltd., Milton, QLD, Australia) MLS which uses a proprietary SLAM algorithm. Each plot was sampled by walking through the vegetation using a rough racetrack or grid pattern (Tiede et al. 2024) which allows for loop closure and adequate sampling density. Figure 3 shows an example of the racetrack pattern used to collect dense MLS point clouds and allows loop closure for SLAM algorithm.

Retro-reflective ground control points were located at the corners and center of each plot and their precise location was surveyed (within 12 mm) using RTK-GPS equipment. The ST-X model uses onboard RTK GNSS which removed the requirement for georeferenced ground points. The 50×50 m pattern took approximately 10 min to sample, and 100×100 m took approximately 20 min. Setup and survey or georeferenced markers took approximately 40 min. Measurements of canopy height and canopy base height were undertaken at five sites to ensure MLS measurements were comparable. Field measurements were undertaken by reading vegetation height against a range pole every meter along a 50-m transect through the center of the plot.

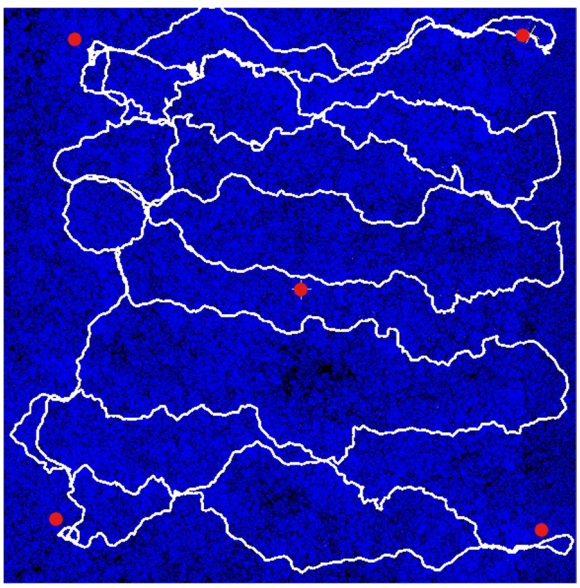


**Fig. 3** Example of racetrack sampling pattern used with backpack mobile laser scanner with discrete points shown in blue, ground control points in red, and the sampling path in white

Raw output was processed using Emesent Aura software to produce discrete-return las point cloud files. Raw file conversion took approximately 10 min for a 0.36-ha plot using a Dell m18 with Intel Core i9-14900HX (2.20 GHz) CPU, 64.0 GB RAM, and 16 GB NVIDIA GeForce RTX4090 GPU. Using Cloud Compare software (Cloud Compare 2024), the point clouds were visually inspected, georeferenced using the ground control points, and height-normalized using a cloth simulation filter (Zhang et al. 2016) with class threshold = 0.1 m, cloth resolution = 1 m, and rigidness = 2. This step took approximately 2 min using the above setup.

The final plot areas varied in size from 0.05 to 0.49 ha with most sites being 0.36 ha. After being georeferenced and height normalized, each plot was divided into a regular grid of subplots of 10 × 10 m as shown in Fig. 4. This resulted in between 6 and 49 subplots per plot. Subplots allowed for distinct areas of variation in metrics to be captured while maintaining appropriate scale of fuel elements with a plot. Initial analysis found that subplots with low point density were significantly different from high-density plots; subplots with fewer than 10,000 returns (i.e., 1000 points/$m^2$) were excluded to ensure metrics were not influenced by low point density. This value was chosen as plots without significant vegetation which were known to be thoroughly sampled still had at least 1000 points/$m^2$.

After filtering, 816 height-normalized 0.01-ha subplots from the 25 sites were used for deriving metrics at both the whole-plot and subplot scale. Calculation of metrics and statistical analysis was undertaken using core functionality within R (R Core Team 2021) unless otherwise noted. Calculation of metrics took approximately 10 min per 0.36-ha plot.

### Selection of fuel metrics

A selection of LiDAR-derived metrics was chosen to represent the structural characteristics and arrangement of fuel in coastal mallee that drive fire behavior. As reported by Telfer et al. (2024), a total of 41 metrics are considered including vegetation height metrics, vegetation cover metrics, strata-based metrics, and metrics commonly used to describe vegetation structure contained in the lidR package (Roussel et al. 2020). A full list of metrics, description, and summary statistics are provided in Additional file 1.

### Estimation of vegetation strata metrics

Current fuel assessment methods used in Australia also rely on strata-based metrics (Hines et al. 2010; DENR 2012). Vegetation strata characteristics were estimated from histogram point density profiles using a method similar to Wilkes et al. (2016). Using the frequency of LiDAR returns with vertical bin size of 0.05 m, a univariate cubic smoothing spline was used to fit a continuous curve to the histogram point density profiles (R Core Team 2021). The smoothing spline was parameterized to provide the best fit without over-smoothing fuel strata nor introducing spurious inflection points (spar = 0.5). The top and bottom height of vegetation strata were estimated using second derivative inflection points of the smoothed spline's fitted values.

$$f''(x_i) \approx f(x_i + 1) - 2f(x_i) + f(x_i - 1) \quad (1)$$

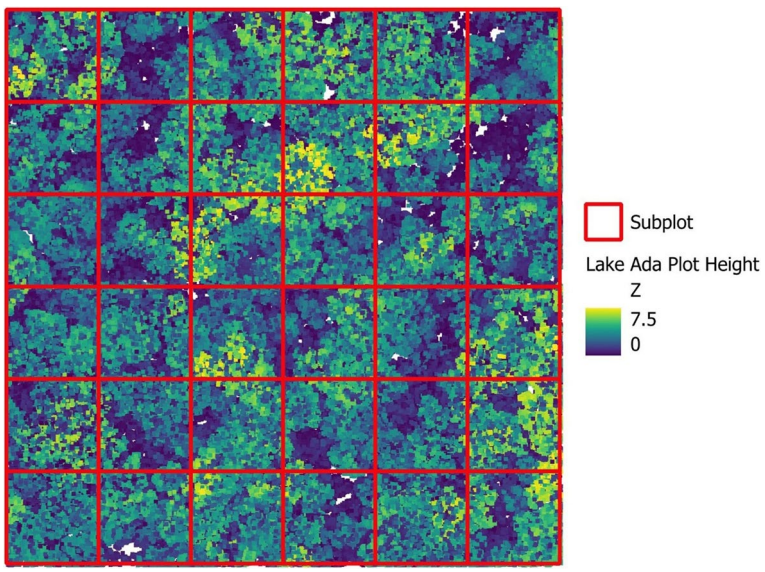



**Fig. 4** Example of subplot sampling for height-normalized georeferenced point cloud for 60 × 60 m plot at Lake Ada site on Kangaroo Island

where $x_i$ the $i$th bin in the LiDAR height distribution, $f(x)$ is the smoothing spline fitted to the binned data, and $f''(x_i)$ is the discrete approximation of the second derivative at $x_i$.

The inflection points were determined by identifying the locations where:

$$h_i : f''(x_i) \times f''(x_i + 1) < 0 \tag{2}$$

where $h_i$ is the height associated with bin $i$. This indicates $h_i$ becomes an inflection point when the second derivative changes sign between $(x_i)$ and $(x_i + 1)$.

Inflection points were labeled as a "top" $(t)$ if the spline is decreasing and "bottom" $(b)$ if the spline is increasing.

$$\ell(p_j) = \begin{cases} t, \frac{df(x_i)}{dx} < 0 \\ b, \frac{df(x_i)}{dx} > 0 \end{cases} \tag{3}$$

where $\ell(p_j)$ is the label (either "top," $t$, or "bottom," $b$) assigned to the $j$th inflection point $p_j$, based on the local slope of the first derivative $f\prime(x_i)$.

When two (or more) consecutive inflection points shared the same label, we merged them by retaining only the point that is most extreme in height for that label

$$P_j^* = \{\{ \begin{matrix} \max\{h_j : \ell(P_j = t)\}, \text{for consecutive top points,} \\ \min\{h_j : \ell(P_j = b)\}, \text{for consecutive bottom points} \end{matrix} \tag{4}$$

where $p_j^*$ denotes the merged inflection points and $h_j$ is the height of $j$th inflection point.

If no "bottom" $(b)$ inflection point existed below the first "top" $(t_1)$, a nominal bottom boundary was defined as a ground level (i.e., 0.0 m). For consistency in calculating canopy metrics, the top height of the tallest stratum was constrained to the 98th percentile height ($z_{98,}$). If the tallest stratum top was within 0.3 m below this limit, it was set to $z_{98,}$; otherwise, it remained unchanged (Eq. 5).

$$t_n = \begin{cases} z_{98,} z_{98} - h(p_{max}) \leq 0.3, \\ h(p_{max}), otherwise. \end{cases} \tag{5}$$

where $t_n$ was the height of the $n$th (tallest) stratum inflection point and $z_{98}$ is the 98th percentile of all points in a subplot.

The depth of each fuel stratum was calculated as the difference in height between pairs of top and bottom inflection points.

$$D_j = t_j - b_j \tag{6}$$

where $D_j$ is the vertical depth and $t_j$ and $b_j$ are the respective top and bottom heights of fuel stratum $j$.

The vertical gap between each fuel stratum was calculated as the height difference between the top of the fuel stratum and the next bottom inflection point.

$$Vgap_j = b_{j+1} - t_j \tag{7}$$

Stratum depth and vertical gaps less than 0.3 m were removed as they were considered to have minimal effect on vertical fire propagation under most sustained fire propagation scenarios and to minimize the possibility of point cloud artifacts introducing spurious fuel strata. After removing inflection points of strata and gaps less than 0.3 m, the depths and gaps were recalculated with the resultant depths and gaps recorded as vertical structure metrics. The maximum strata depth ($D_{max}$) and gap ($Vgap_{max}$) of each subplot were also considered relevant to fire propagation and were recorded for each subplot. Examples of the variation in coastal mallee strata are demonstrated in Fig. 5 which shows the vegetation histogram profile (Fig. 5a, c) and an extract of the point cloud profile along a transect through the center of the corresponding subplot (Fig. 5b, d). The top height of strata is shown as red dashed horizontal lines, and the bottom strata heights are blue horizontal lines and the 98th percentile height is shown in orange. The very lowest layer of voxels is excluded from strata calculations and figures to avoid misrepresenting low, unburnable features as vegetation due to limitations in the cloth simulation filter method.

#### Derivation of horizontal fuel metrics

Horizontal fuel connectivity was estimated using point clouds voxelized at 0.25 $m^3$ resolution using total vegetation cover ($cov_t$) (Eq. 8) and overstory cover ($cov_o$) (Eq. 9).

$$cov_t = \frac{N_{0.5} \times v^2}{A} \tag{8}$$

where $N_{0.5}$ is the number of voxels above 0.5 m, $v$ is the resolution of the voxels (0.25 m), and $A$ is the area of the plot. The 0.5-m threshold was used to ensure no ground returns or rocks were included as vegetation returns.

$$cov_o = \frac{N_{cbh} \times v^2}{A} \tag{9}$$

where $N_{cbh}$ is the number of voxels above canopy base height, $v$ is the resolution of the voxels (0.25 m), and $A$ is the area of the plot.

#### Derivation of height and vertical structure fuel metrics

Vertical fuel structure was estimated using a range of existing LiDAR-derived structure metrics. The lidR package (Roussel et al. 2020) was used to estimate maximum ($Z_{max}$), mean ($Z_{mean}$), 98th percentile ($Z_{98}$), standard deviation ($Z_{SD}$), and skewness ($Z_{skew}$) of height, and vertical complexity index (VCI) (van Ewijk et al. 2011). Canopy base height (CBH) was estimated using the bottom height of the tallest stratum. Gap fraction profile ($GF_P$) was calculated at

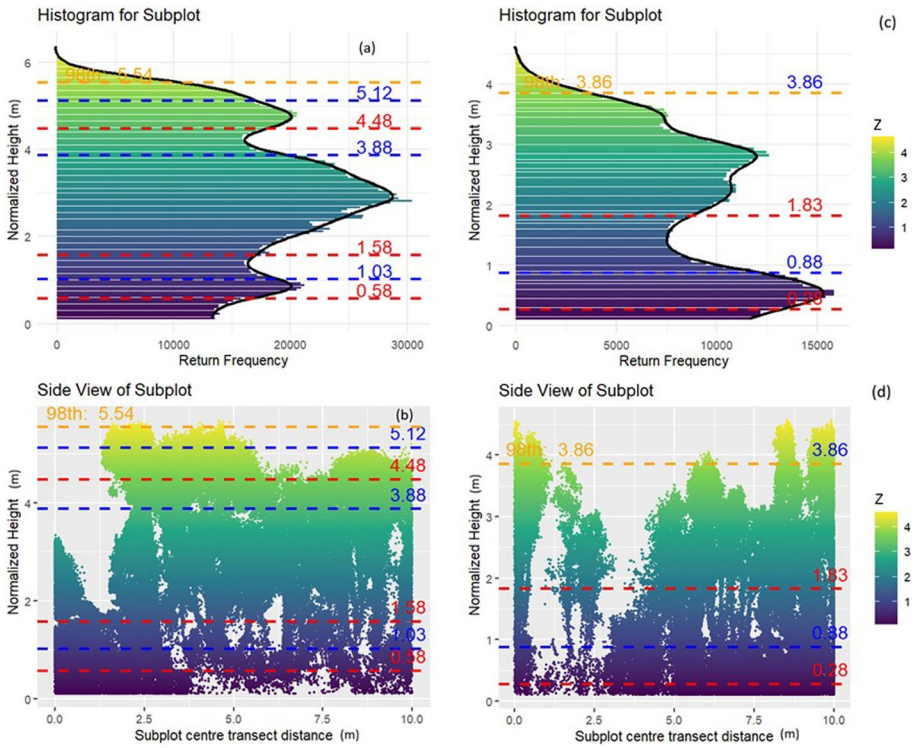



**Fig. 5** Example vegetation profiles showing the different strata and their corresponding height boundaries (measured in meters), with the red line indicating the bottom boundary of a stratum and the blue line indicating the top boundary of a stratum. The orange line represents the 98th percentile height of all points

1 m above ground and at the midpoint of each fuel strata ($GF_1$ and $GF_2$) using 0.5 m vertical resolution (Bouvier et al. 2015). Canopy relief ratio (CRR), foliar height diversity (FHD), and vertical distribution ratio (VDR) were estimated in a similar manner to Ross et al. (2024) using the voxelized point clouds at 0.25 $m^3$ resolution.

### Statistical analysis of metrics

Normality of derived metrics was determined using histograms and Q-Q plots, together with skewness, kurtosis, and Shapiro–Wilk tests. The difference between the whole plot and subplots was examined by comparing the mean and variance using Welch's *t*-test and Fligner–Killeen test. These tests were chosen over standard *t*-test and *F*-tests as the sample size of the two methods was not even, nor normally distributed. Spearman's rank correlation coefficients were computed to assess the monotonic relationships among the fuel metrics. The non-parametric Kruskal–Wallis rank sum test was used to determine whether there are statistically significant differences between the medians of each site on Kangaroo Island and Eyre Peninsula and between the age-rainfall classes. Given the exploratory nature of metric screening, no correction for multiple comparisons was applied.

## Results

### Summary statistics of plots

The average sample point density of all MLS point clouds was 15,617 points/$m^2$ with a range of 3168–42,129 points/$m^2$, depending on vegetation density and sample dwell time. Table 1 shows the point density alongside plot sizes, number of 100 $m^2$ sublots, the years since fire, average annual rainfall, and region of each site. The plots were spread evenly across Kangaroo Island (KI) and Eyre Peninsula (EP) although the majority of sites were less than 30 years since last fire and more than 500 mm average annual rainfall. There was no correlation between point density and the size of the plot, age (years since fire), rainfall, or region. Summary statistics of each metric per plot is provided in Additional file 2.

### Fuel strata metrics

The descriptive statistics and intercomparison test results based on spatial scale (whole-plot and subplot scale) for the fuel strata-based metrics are presented in Table 2. While the mean of the height of the lowest stratum ($Zmax_1$) did not differ by spatial scale, the means of the second-lowest stratum ($Zmax_2$), the gap between the lowest and second-lowest strata ($Vgap_{1_2}$), and the largest gap between any strata ($Vgap_{max}$) significantly

**Table 1** Summary statistics of subplot, LiDAR sample point density, years since fire, and average annual rainfall for coastal mallee plots on Eyre Peninsula (EP) and Kangaroo Island (KI)

| Plot ID | Plot area ($m^2$) | Subplots | Density ($p/m^2$) | Age (years since fire) | Rainfall (mm) | Region |
|---|---|---|---|---|---|---|
| 1 | 3600 | 36 | 16,065 | 22 | 500 | EP |
| 2 | 2500 | 25 | 8155 | 22 | 500 | EP |
| 3 | 3600 | 36 | 14,724 | 22 | 500 | EP |
| 4 | 4900 | 49 | 7717 | 22 | 500 | EP |
| 5 | 3600 | 36 | 3168 | 22 | 500 | EP |
| 6 | 3600 | 36 | 7358 | 22 | 500 | EP |
| 7 | 2500 | 25 | 17,560 | 22 | 500 | EP |
| 8 | 2500 | 25 | 22,747 | 17 | 550 | KI |
| 9 | 2500 | 25 | 31,631 | 17 | 550 | KI |
| 10 | 3600 | 36 | 22,962 | 17 | 550 | KI |
| 11 | 1000 | 10 | 14,580 | 17 | 550 | KI |
| 12 | 600 | 6 | 42,129 | 16 | 550 | KI |
| 13 | 2500 | 25 | 18,258 | 15 | 550 | KI |
| 14 | 4900 | 49 | 13,262 | 16 | 550 | KI |
| 15 | 4900 | 49 | 5741 | 16 | 550 | KI |
| 16 | 4900 | 49 | 8361 | 16 | 550 | KI |
| 17 | 2500 | 25 | 7660 | 14 | 500 | EP |
| 18 | 2500 | 25 | 14,264 | 45 | 550 | KI |
| 19 | 3600 | 36 | 12,171 | 16 | 550 | KI |
| 20 | 3600 | 36 | 15,816 | 16 | 550 | KI |
| 21 | 3600 | 36 | 22,911 | 16 | 550 | KI |
| 22 | 4900 | 49 | 7562 | 19 | 475 | EP |
| 23 | 3600 | 36 | 12,761 | 19 | 475 | EP |
| 24 | 3600 | 36 | 19,776 | 23 | 460 | EP |
| 25 | 3600 | 36 | 23,076 | 57 | 460 | EP |

**Table 2** Comparison of LiDAR-derived fuel strata metrics in all coastal mallee sites for whole plots and average of subplot samples using Welch's *t*-test and Fligner–Killeen (FK) test of variance. Data are presented as mean ± SE with variance in parentheses

| Metric | Whole plot | Subplot | Welch's *t*-test | | | FK test | | |
|---|---|---|---|---|---|---|---|---|
| | Mean | Mean | *t* | df | *P* | $X^2$ | df | *P* |
| $strata_n$ | 2.08 ± 0.08 (0.16) | 2.38 ± 0.03 (0.55) | −3.54 | 29.36 | 0.001* | 6.88 | 1 | 0.009* |
| $Zmax_1$ (m) | 0.98 ± 0.08 (0.17) | 1.12 ± 0.02 (0.39) | −1.54 | 27.51 | 0.134 | 2.85 | 1 | 0.092 |
| $Zmax_2$ (m) | 3.42 ± 0.16 (0.6) | 3.01 ± 0.03 (0.76) | 2.56 | 24.87 | 0.017* | 2.2 | 1 | 0.138 |
| $Vgap_{1_2}$ (m) | 1.1 ± 0.8 (0.14) | 0.76 ± 0.01 (0.12) | 4.45 | 24.3 | <0.001* | 1.6 | 1 | 0.206 |
| $Vgap_{max}$ (m) | 1.14 ± 0.07 (0.11) | 0.81 ± 0.01 (0.11) | 4.9 | 24.55 | <0.001* | 0.17 | 1 | 0.677 |
| $D_{max}$ (m) | 1.64 ± 0.11 (0.3) | 1.61 ± 0.02 (0.37) | 0.28 | 25.84 | 0.783 | 0.01 | 1 | 0.913 |

*denotes significant test result ($p < 0.05$) for corresponding Welch's *t*-test or Fligner–Killeen test

differed between whole plot and subplots. The means of strata depth metrics were not significantly different between whole-plot or subplot scales. The variance of all the fuel strata metrics was only significantly different for number of strata ($strata_n$).

Height and vertical gap of the third stratum and above did not have enough samples to be meaningful as the whole-plot method only had three sites with three strata detected using the inflection point methods, while the maximum number of strata detected using the smaller subplots was five. This suggests the 10 × 10 m subplots may be better at detecting variability in number, height, and connectivity of fuel strata in coastal mallee.

Using the subplot data, the maximum height of each stratum was strongly correlated with the corresponding strata depth ($r=0.9$ for $Zmax_1$ and $r=0.75$ for $Zmax_2$). $Vgap_{max}$ was also strongly correlated to $Vgap_{1_2}$ ($r=0.85$). The results of the Kruskal–Wallis test indicate that the median values of all the strata metrics are different between the age-rainfall classes ($P \leq 0.05$). However, the median depth and number of strata were not different between the two broad geographic regions (Kangaroo Island and Eyre Peninsula).

Based on these results, $Zmax_1$ and $Zmax_2$ (mean height of 1.12 m ± 0.02 and 3.01 m ± 0.03, respectively) and the $Vgap_{max}$ (mean gap 0.81 m ± 0.01) are the most useful strata metrics in describing the structure of coastal mallee fuel. These three metrics were not strongly correlated with any of the horizontal or vertical structure metrics.

### Horizontal fuel metrics

The mean vegetation cover was significantly higher using the subplot method compared with a single cover number for the whole plot (Table 3). The variance of both the vegetation cover and overstory cover metrics was also significantly higher using the subplot method.

$Cov_t$ and $Cov_o$ were not correlated ($r=0.08$). The median values of both cover metrics were different between the age-rainfall classes; however, only overstory cover differed between the geographic regions. The variability of $Cov_t$ above 0.5 m is lower than $Cov_o$ and 16% of subplots have 100% cover above 0.5 m. This limits the usefulness of $Cov_t$ as a distinguishing metric. $Cov_o$ is the preferred cover metric for this coastal mallee dataset and has a mean value of 0.4 ± 0.008. Overstory cover is not correlated with other strata or vertical structure metrics.

### Vertical fuel metrics

Vertical fuel metrics results include height, gap fraction, and vertical distribution metrics. The descriptive statistics for vertical fuel metrics are listed in Table 4. The mean and variance of height metrics were not significantly different except for the maximum height of returns ($Z_{max}$). This is expected as $Z_{max}$ is affected by extremes in small amounts of vegetation which may protrude from an otherwise lower canopy.

While the mean of the profile gap fraction was not significantly different between the whole-plot and subplot methods, the gap fraction for the lowest two strata was lower using the subplot method. This is possibly because the smaller plot size had more tightly defined upper and lower bounds of strata and therefore more likely to be filled with vegetation (i.e., lower gap fraction) than the whole plot which would include more gaps within the strata due to the coarser strata definition of larger plots.

The means of all vertical distribution metrics are different between the whole-plot and subplot methods. $Z_{skew}$, CRR, VCI, VDR, and FHD are sensitive to detail in the vertical distribution of fuel and it is expected that these metrics would differ between subplots and full plots due to the averaging of vertical point distribution in full plots. Only CBH, the gap fraction metrics ($GF_P$, $GF_1$, $GF_2$), $Z_{skew}$, and VDR showed a difference in variance between whole plot and subplots.

Maximum, mean, and standard deviation heights ($Z_{max}$, $Z_{mean}$, $Z_{SD}$) were all strongly correlated ($r \geq 0.9$) with the 98th percentile height ($Z_{98}$) using whole-plot and subplot methods. 98th percentile height is preferred as it represents canopy height without being affected by occasional taller parts of the canopy. FHD was also very strongly correlated with height metrics ($r=0.98$) using both whole-plot and subplot methods.

Gap fractions of the lowest two strata ($GF_1$ and $GF_2$) were moderately correlated with each other ($r=0.52$), but not with the gap fraction of the whole vegetation profile ($GF_P$). The median values of all three gap fraction metrics differed between age-rainfall classes. $GF_1$ and $GF_2$ differed between geographic regions but $GF_P$ did not. However, given $GF_P$ is not correlated with other metrics and mean $GF_P$ is not affected by plot size, it is included as a candidate metric to describe vertical structure alongside $GF_1$ and $GF_2$.

VDR, CRR, and $Z_{skew}$ are strongly correlated which is expected as they each measure the amount of vegetation over the distribution of height. The Kruskal–Wallis rank sum test between age-rainfall classes showed that median values of all three metrics differed between classes, but $Z_{skew}$ showed the most significant difference between geographic regions ($P \leq 0.001$), CRR was marginally significant (0.05), and VDR did not differ between regions

**Table 3** Comparison of horizontal cover fuel metrics in all coastal mallee sites for whole plots and average of subplot samples using Welch's *t*-test and Fligner–Killeen (FK) test of variance. Data are presented as mean ± SE with variance in parentheses

| Metric | Whole plot | Subplot | Welch's *t*-test | | | FK test | | |
|---|---|---|---|---|---|---|---|---|
| | Mean | Mean | *t* | df | *P* | $X^2$ | df | *P* |
| $Cov_t$ (%) | 0.89 ± 0.11 (0.02) | 0.93 ± 0.03 (0.07) | −3.32 | 28.16 | 0.003* | 4.09 | 1 | 0.043* |
| $Cov_o$ (%) | 0.40 ± 0.32 (0.25) | 0.40 ± 0.08 (0.5) | −0.04 | 26.99 | 0.965 | 5.24 | 1 | 0.022* |

*denotes significant test result ($p$ <0.05) for corresponding Welch's *t*-test or Fligner–Killeen test

**Table 4** Comparison of vertical fuel metrics in all coastal mallee sites for whole plots and average of subplot samples using Welch's *t*-test and Fligner–Killeen (FK) test of variance. Data are presented as mean ± SE with variance in parentheses

| Metric | Whole plot | Subplot | Welch's *t*-test | | | FK test | | |
|---|---|---|---|---|---|---|---|---|
| | Mean | Mean | *t* | df | *P* | $X^2$ | df | *P* |
| $Z_{98}$ (m) | 4.21 ± 0.16 (0.67) | 3.95 ± 0.03 (0.92) | 1.53 | 26.08 | 0.139 | 1.82 | 1 | 0.177 |
| $Z_{max}$ (m) | 6.34 ± 0.24 (1.44) | 4.93 ± 0.04 (1.29) | 5.82 | 25.34 | <0.001* | 0 | 1 | 0.993 |
| $Z_{mean}$ (m) | 1.95 ± 0.07 (0.14) | 1.92 ± 0.02 (0.23) | 0.41 | 26.51 | 0.688 | 3.51 | 1 | 0.061 |
| $Z_{SD}$ (m) | 1.16 ± 0.05 (0.06) | 1.09 ± 0.01 (0.08) | 1.41 | 26.23 | 0.17 | 1.57 | 1 | 0.21 |
| CBH (m) | 2.13 ± 0.14 (0.46) | 2.26 ± 0.04 (1.15) | −0.95 | 27.85 | 0.351 | 6.93 | 1 | 0.008* |
| $GF_P$ | 0.81 ± 0.01 (0.001) | 0.8 ± 0.002 (0.003) | 1.74 | 30.95 | 0.092 | 8.19 | 1 | 0.004* |
| $GF_1$ | 0.83 ± 0.01 (0.01) | 0.69 ± 0.004 (0.01) | 9.59 | 28.01 | <0.001* | 6.51 | 1 | 0.011* |
| $GF_2$ | 0.97 ± 0.01 (0.002) | 0.89 ± 0.003 (0.01) | 7.95 | 28.2 | <0.001* | 14.46 | 1 | <0.001* |
| $Z_{skew}$ | 0.23 ± 0.03 (0.03) | 0.14 ± 0.01 (0.08) | 2.63 | 28.78 | 0.014* | 7.16 | 1 | 0.007* |
| CRR | 3.27 ± 0.08 (0.15) | 2.59 ± 0.01 (0.14) | 8.76 | 25.42 | <0.001* | 0.16 | 1 | 0.686 |
| VCI | 0.77 ± 0.01 (0.002) | 0.83 ± 0.01 (0.001) | −5.76 | 17.48 | <0.001* | 2.72 | 1 | 0.099 |
| VDR | 0.31 ± 0.01 (0.002) | 0.39 ± 0.01 (0.01) | −10.16 | 29.14 | <0.001* | 9.54 | 1 | 0.002* |
| FHD | 2.82 ± 0.04 (0.04) | 2.72 ± 0.01 (0.06) | 2.45 | 26.36 | 0.021* | 2.74 | 1 | 0.098 |

*denotes significant test result ($p$ <0.05) for corresponding Welch's *t*-test or Fligner–Killeen test

($P$=0.49). VCI differed between age-rainfall classes but was not significant between locations ($P$=0.083).

A scatterplot matrix of the refined list of fuel metrics is presented in Fig. 6 showing only moderate positive correlation between $Z_{max1}$ and $Z_{max2}$ ($r$=0.53), $GF_1$ and $GF_2$ ($r$=0.52), and between $Z_{skew}$ and $GF_2$ ($r$=0.61). VCI is moderately positively correlated with height ($Z_{98}$) and negatively correlated with $GF_1$, $GF_2$, CRR, VDR, and FHD.

## Discussion

The intent of this study was to identify objective, repeatable methods for quantifying horizontal and vertical fuel connectivity in coastal mallee as an exemplar of complex shrubland ecosystems. The aim was to refine the list of candidate LiDAR metrics for shrubland fuel modeling rather than to directly model fire behavior. While the methods provided guidance on metrics for distinguishing broad fuel classes, there is an untested assumption that fire behavior varies among these classes. However, the methods and results presented provide a template for capturing and processing LiDAR point clouds to derive quantified fuel-structure metrics. There are other methodological choices and assumptions that are discussed below and warrant further research in the future.

### Comparing LiDAR-derived metrics at different spatial scales

Calculating fuel structure metrics at a 10 m × 10 m subplot retains a wider range of values than whole plots. This suggests using smaller subplots may quantify variability in horizontal fuel connectivity, which is smoothed out by the calculated metrics for a larger area. Overall, the greater variability of most metrics obtained from the subplot method, compared to those calculated using the whole plot, suggests that calculating metrics on a subplot scale is warranted. A wider range of values is useful for detecting subtle variations in fuel structure, which may account for differences in fire behavior reported by fire managers. While 10 × 10 m subplot scale was chosen for the MLS-derived point clouds in this study, it is somewhat arbitrary and testing different resolutions or methods of segmenting metrics into relevant features (e.g., clusters of vegetation or individual trees/shrubs) may further refine the utility of LiDAR-derived metrics in shrublands (Taneja et al. 2021). Metrics calculated at a tree or clump scale may provide useful insight; however, distinguishing

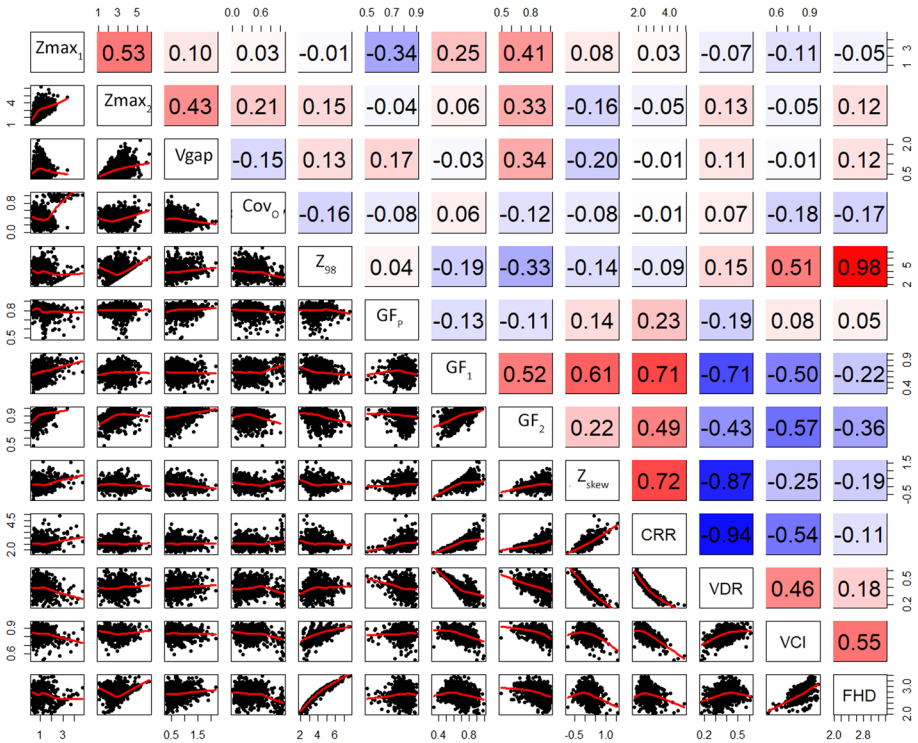



**Fig. 6** Correlation matrix of LiDAR metrics in coastal mallee including scatterplot with loess fit trendline (lower left) and Spearman correlation coefficient (upper right)

individual trees or vegetation patches has proved difficult in complex shrublands such as coastal mallee as most methods have been developed for taller forests with distinct canopies (Viedma et al. 2024; Almeida et al. 2019).

Accounting for variability in fuel structure is important for fires at the prescribed burn scale under milder conditions. Prescribed fires are generally able to be applied in a controlled manner and fire behavior is more sensitive to structural changes than in a wildfire context. Based on a combination of variability, low correlation with simpler metrics and ability to distinguish between age-rainfall classes (as a proxy for fuel structure), max vertical strata gap ($Vgap_{max}$), the height of the lowest two strata ($Z_{max1}$, $Z_{max2}$), gap fraction metrics ($GF_P$, $GF_1$, $GF_2$), height skewness ($Z_{skew}$), and vertical distribution ratio (VDR) all warrant further investigation into the correlation to fire behavior at the prescribed burn scale. At a larger spatial scale and increased fire behavior, the effect and importance of fuel on fire spread has been shown to decrease (Cruz et al. 2022). The need for fine-scale fuel metrics diminishes, and coarser approximations of fuel structure may be more suitable. Our results suggest that canopy height, canopy cover, and the whole vegetation profile gap fraction are less sensitive to variations due to plot size, potentially making them more suitable for wildfire-scale modeling.

### LiDAR fuel metrics for wildfire modeling

Vertical structure metrics were generally correlated with each other and had varying ability to distinguish between locations and age-rainfall classes. Given that canopy height is the most commonly used fuel metric in shrubland fire models (Telfer et al. 2024) and is reliably derived from LiDAR point clouds, height is the preferred metric over strongly correlated alternatives such as FHD. The ultimate distinction from a fuel perspective is in their ability to distinguish fire behavior. Future coastal mallee fire behavior studies could consider a range of strata-based (height of strata, gap fraction within strata, and vertical gap between strata), voxelized canopy cover, canopy height, and vertical fuel structure metrics (VCI and either $Z_{skew}$, CRR, or VDR).

The two fuel inputs used in the semi-arid heath and the shrubland fire spread models (Cruz et al. 2013; Anderson et al. 2015), canopy height and overstory cover, had mean values of 3.9 m and 40%, respectively, across all sampled coastal mallee sites. The overstory cover results of coastal mallee are beyond the bounds of the empirical semi-arid mallee model, which suggest it may not be suitable at predicting fire spread in coastal mallee (Cruz et al. 2013). Mean vegetation height was within the bounds of

the semi-arid mallee model and at the upper end of the shrubland model. This suggests the shrubland model may be suitable for predicting rate of spread of fires in coastal mallee using LiDAR-derived fuel height measurements collected with MLS. The shrubland model lacks the spread probability component to predict go/no-go conditions for prescribed and wildfires, so future research should investigate the relationship between our recommended LiDAR structure metrics and fire propagation in coastal mallee. In particular, we recommend LiDAR-derived vertical gap and complexity metrics be considered in future fire spread research in shrublands.

Coastal mallee shrublands have a dense, complex fuel structure similar to other Mediterranean shrublands. The LiDAR-derived metrics used in this study could also be applied to other fire-prone shrublands around the world to quantify fuel structure. As most previous studies have focused canopy height and biomass estimations, methods to quantify vertical distribution of fuel and strata may provide new insights into fire propagation in shrublands.

#### Strengths and limitations of MLS methods

LiDAR-derived fuel metrics are widely recognized as proxies for many field-based wildfire fuel measurement methods (Alonso-Rego et al. 2020; Hillman et al. 2019; Vierling et al. 2013). LiDAR methods offer several advantages including measurement precision and accuracy, reduced field time, the ability to reprocess point clouds using new techniques, and reduced observer bias compared with visual field-based methods commonly used in Australia (Volkova et al. 2016; Spits et al. 2017). Proxies for fuel continuity, such as those described in Ross et al. (2024), are not easily measured using traditional field-based methods and would be better estimated using detailed point cloud datasets. MLS using SLAM algorithms offers practical advantages over TLS in setup time and required expertise to capture and process point clouds, while producing fuel metrics with comparable accuracy and precision for many structural fuel measurements (Vandendaele et al. 2022). MLS may reduce obscuration of subcanopy features relative to TLS where limited scan positions in dense environments can lead to local occlusion (Donager et al. 2021), and relative to ALS, as the operator, can navigate beneath the canopy and around obstacles during data acquisition (Donager et al. 2021; Hoffrén et al. 2024; Tiede et al. 2024). However, in shrublands, physical access to some areas of dense vegetation in coastal mallee proved difficult, resulting in lower point densities. Variations in point densities may affect strata definitions and strata-based metrics, with lower density reducing the ability to resolve fine vertical structure and number of strata. In this study, the influence of low point density was minimized by applying a minimum threshold during the voxelization process. Voxel-based methods are less prone to variation in point densities due to the aggregation of points, which should be a consideration in future research.

MLS remains limited in its ability to distinguish litter fuel metrics due to difficulty separating leaf litter from non-flammable surface elements such as bare earth and rocks (Abdollahi and Yebra 2025). However, regression models linking LiDAR-derived structural metrics with field observations have been used to predict surface fuel load (Lin et al. 2024; Holt, Marre, and Telfer 2021).

#### Influence of other method choices on MLS-derived fuel metrics

The CSF method requires parameterization to maximize accuracy, and small errors in the modeled ground surface have a relatively large impact on litter fuel metrics (Hillman et al. 2021a, 2021b, 2021c; Fernández-Guisuraga et al. 2025). Colorized point clouds could improve the ability to distinguish between ground and vegetation, although still possibly limited in soil and litter or dead shrubs that are a similar tone. Initial trials of colorized point clouds using the Hovermap system show variable results in the accuracy of color point mapping.

The coastal mallee plots were divided into 10×10 m subplots for consistency with accepted fuel collection methods and anticipated influence of fuel characteristics at prescribed burn scale. However, the effect of the arbitrary size and arrangement of subplots was not tested by this research and the effect on fuel parameters is not known. Alternatives to arbitrary plot size and layout could use a natural clustering method to assess metrics in clusters of discrete fuel arrays (Bradstock and Gill 1993). Future studies could compare the approach used here with different size and method of determining subplots.

We relied on threshold and voxel sizes from similar studies and knowledge of fire behavior propagation (Ross et al. 2022, 2024) with minimal sensitivity testing of metrics to voxel and thresholds. As voxel size has been shown to influence connectivity metrics in shrublands (Homainejad et al. 2022; Ross et al. 2022), future studies could look at the sensitivity of metrics derived from high-density MLS point clouds to voxel size and above-ground thresholds.

## Conclusion

LiDAR-derived metrics collected by MLS can be used to quantify fuel structure of coastal mallee shrublands. A combination of strata, canopy cover, height, and vertical structure metrics described a range of fuel attributes and can distinguish between age-rainfall classes.

MLS methods offer improved fidelity and less effort than traditional field-based methods. In the future,

quantifying changes in fuel structure could help determine critical fire behavior thresholds and explain anecdotal differences in fire spread rates and the go/no-go phenomenon observed in shrublands.

Using coastal mallee as an exemplar of complex fuel structure, MLS-derived fuel metrics could also be applied to analogous vegetation types in similar ecosystems, where fuel connectivity is a key driver of fire behavior. This could be particularly important for modeling fires at low to moderate intensity, where fuel structure is more important than higher-intensity wildfires (Cruz et al. 2021, 2022).

The ultimate test of proposed fuel metrics is not only the ability to determine fuel classes, but also correlation with fire behavior. Future field studies on the correlation between MLS-derived fuel metrics and fire behavior could enable the development of empirical and physical fire models using detailed fuel-structure metrics.

## Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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**Authors' contributions**
All authors identified the questions and designed the study. ST conducted the field work, collated and analysed the data and wrote the manuscript. All authors contributed to editing and revising of the manuscript.

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**Data availability**
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Ethics approval and consent to participate**
Not applicable.

**Consent for publication**
Not applicable.

**Competing interests**
The authors declare no competing interests.

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